Product Specification & Requirements Document (PRD) — v1.0 · 16 July 2026

TEK Restaurant Intelligence Platform

Client	TEK — Tasty E-Kitchen Ltd, London, UK (tekvers.ai)
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Version	2.0 — Revised all-inclusive proposal
Status	For client review

1. Executive Summary

TEK positions itself as “the intelligence layer that powers the future of hospitality.” This document specifies the buildout of the core platform that delivers that promise: a multi-tenant SaaS for independent restaurants and hospitality groups, built from scratch, comprising five integrated modules:

1. **First-party customer food-preference database** — TEK-owned capture surfaces that turn every guest interaction into structured preference data.
2. **AI meal recommendation engine** — personalised dish recommendations per guest, per venue, with human-readable reasoning.
3. **CRM with automated personalised marketing and loyalty programmes** — segmentation, automated campaigns (email/SMS), and a configurable loyalty engine.
4. **AI predictive analytics** — churn risk, demand forecasting, and menu-performance intelligence surfaced in a venue dashboard.
5. **TEK Guest mobile app** — iOS & Android app (single cross-platform codebase) bringing the menu, ordering, personalised recommendations, loyalty wallet and consent-gated push notifications to guests’ phones.

The platform is delivered in five fixed-price phases, each independently valuable and demonstrable, quoted as a single **all-inclusive engagement** (Section 12).

Out of scope for this engagement: VR food visualisation/commerce and hardware/smart-kitchen infrastructure. The architecture leaves clean seams for both.

2. Background, Problem & Goals

2.1 Problem

Independent restaurants lose the retention game: the guest relationship is owned by aggregators and generic booking tools. Venues rarely know *who* their guests are, *what* they like, or *when* they are about to lapse. TEK's own market framing cites customer-retention struggles affecting 86% of hospitality businesses.

2.2 Product goals

- **G1** — Give every TEK venue a first-party, GDPR-compliant guest database it owns, populated automatically from TEK capture surfaces.
- **G2** — Convert that data into measurable revenue actions: recommendations that lift average order value, campaigns that lift visit frequency, loyalty that lifts retention.
- **G3** — Give operators forward-looking intelligence (churn, demand, menu performance) rather than backward-looking reports.
- **G4** — Support TEK's commercial model (Starter Intelligence at £249/venue/month; Enterprise custom) with per-venue multi-tenancy, seat management, and usage isolation from day one.

2.3 Success criteria (platform-level)

Metric	Target (first 6 months post-launch)
Guest profiles captured per active venue	≥ 500
Guests with ≥ 3 preference signals	≥ 60% of profiles
Recommendation click-through (QR menu)	≥ 15%
Campaign-attributed repeat visits	measurable per campaign, ≥ 8% redemption
Churn-model precision @ top decile	≥ 2× baseline repeat rate
Dashboard weekly active usage	≥ 70% of paying venues

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3. Scope

3.1 In scope

- Multi-tenant SaaS core: organisations → venues → seats (role-based access), venue onboarding wizard.
- Guest-facing capture surfaces: QR web menu, web ordering page, signup/consent forms, post-visit feedback prompts, loyalty join flow.
- Customer food-preference database (Module 1) with a unified guest-event spine.
- AI meal recommendation engine (Module 2).
- CRM, campaign automation (email + SMS), and loyalty engine (Module 3).
- Predictive analytics: churn scoring, demand forecasting, menu-performance analytics (Module 4).
- Venue-facing business intelligence dashboard.
- GDPR/UK-DPA compliance tooling: consent records, export, right-to-erasure.
- TEK internal admin panel (tenant management, plan limits, platform health).
- TEK Guest mobile app for iOS & Android (Module 5) with push notifications.

3.2 Out of scope (this engagement)

- VR/3D food visualisation and VR commerce.
- Payments/ordering fulfilment beyond capture (no kitchen display, no delivery logistics).
- Multi-language localisation (English-first; i18n-ready string handling).
- Per-venue custom-branded mobile apps, offline mode, and in-app payment processing (the TEK Guest app ships under the TEK brand; venues appear within it).

3.3 Assumptions

- TEK supplies brand assets, domain(s), and legal copy (privacy policy, terms) — templates provided by OEFR.
- TEK supplies Apple Developer and Google Play accounts (and store listing approvals) for the mobile app; OEFR prepares builds and store assets.
- TEK operates the commercial relationships with venues; OEFR delivers the platform.
- Third-party service costs (hosting, email/SMS delivery, LLM API usage) are billed directly to TEK's accounts; estimates in Section 11.
- Pilot cohort of 3–10 venues available for Phase 2–3 validation.

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4. Users & Personas

Persona	Description	Primary needs
Venue Owner/GM	Runs 1–3 sites, time-poor	Retention, revenue lift, zero-admin tooling
Marketing Manager	Group/franchise level	Segments, campaigns, loyalty configuration, attribution
Front-of-house staff	Service staff	Guest preference card at a glance (allergies, favourites, VIP)
Guest (diner)	End customer	Fast menu/ordering, relevant recommendations, rewards worth joining, control over their data
TEK Admin	TEK's own team	Tenant provisioning, plan enforcement, platform health, model performance

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5. Module 1 — First-Party Customer Food-Preference Database

The foundation. Every other module reads from this.

5.1 Concept: the guest-event spine

All guest activity is written as immutable events against a guest profile: `menu_view`, `dish_view`, `order_placed`, `order_item`, `feedback_given`, `campaign_open`, `campaign_click`, `reward_redeemed`, `visit_checkin`, `preference_declared`. Preferences are **derived** (from behaviour) and **declared** (explicitly stated), kept distinct and both queryable.

5.2 Capture surfaces (v1, TEK-owned)

Surface	Captures
QR web menu (per venue/table)	Menu/dish views, dwell, session device link
Web ordering page	Orders, items, modifiers, spend, time-of-day
Signup & loyalty join flow	Identity (email/phone), declared preferences, dietary needs, marketing consent
Post-visit feedback prompt (email/SMS link)	Dish ratings, visit rating, free-text comment
Staff quick-note (dashboard)	FOH-entered notes: allergies, occasions, VIP flags

5.3 Guest profile

- Identity: email and/or mobile (E.164), name optional; anonymous sessions merge into a profile on identification (deterministic matching only).
- Declared: dietary requirements (vegan, halal, gluten-free, allergens list), cuisine likes/dislikes, spice tolerance, favourite dishes, occasions (birthday, anniversary).
- Derived: taste vector (per taxonomy tags), price-band affinity, visit cadence, channel preference, RFM scores (recency/frequency/monetary).
- Consent object: per-channel marketing consent with timestamped audit trail; profiling opt-out flag honoured by Modules 2–4.

5.4 Menu & taxonomy

- Venue menu manager: categories, dishes, prices, photos, availability windows.
- TEK food taxonomy: every dish tagged (cuisine, ingredients, allergens, spice level, dietary flags, preparation) — tags proposed automatically by LLM from the dish name/description, confirmed by the venue in onboarding.

5.5 Acceptance criteria (Module 1)

- A guest can scan a QR code, browse the venue menu, and their session events persist and merge into their profile after signup.
- A venue can view, search, filter, and export its guest list; guests are strictly invisible across tenants.
- A guest can request their data (export) and erasure; erasure completes across all modules within 30 days and is logged.
- Preference records distinguish declared vs derived, each with provenance and timestamps.

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6. Module 2 — AI Meal Recommendation Engine

6.1 Behaviour

- **Known guest, on QR menu/ordering page:** “For you” rail of 3–6 dishes ranked by taste-vector match, order history, dietary constraints (hard filters), availability, and margin weighting (venue-configurable).
- **Unknown guest:** venue-level popularity + time-of-day priors (cold start), improving within the session from dish views.
- **Explanations:** each recommendation carries a short human-readable reason (“Because you loved the Jollof Special”), LLM-generated from structured signals — never invented facts.
- **Staff view:** FOH preference card shows the same top picks for table-side upsell.

6.2 Engine design

- Stage 1 (retrieval): hard constraint filtering — allergens, dietary flags, availability. **Allergen safety is absolute: a dish containing a declared allergen is never recommended, and the filter is enforced in code, not by the model.**
- Stage 2 (ranking): weighted hybrid — item-based collaborative filtering on order events + content similarity on taxonomy tags + recency/novelty terms. Deterministic, unit-testable scoring.
- Stage 3 (presentation): LLM formats explanation copy from the scoring evidence.
- Feedback loop: impressions and clicks logged as events; weekly offline evaluation (CTR, add-to-order rate) per venue.

6.3 Acceptance criteria (Module 2)

- Recommendations respond < 300 ms p95 (excluding explanation copy, which streams after).
- A guest with a declared allergen never sees a dish containing it (verified by automated test across full menu permutations).
- Venue can toggle margin weighting and see recommendation CTR in the dashboard.
- Cold-start venues get sensible popularity-based rails from day one.

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7. Module 3 — CRM, Automated Marketing & Loyalty

7.1 CRM & segmentation

- Segment builder over profile + event data: attributes (dietary, preferences, RFM band, churn-risk band from Module 4), behaviours ("ordered ≥ 2× in 30 days", "viewed menu but didn't order"), and consent state (always enforced).
- Saved segments update dynamically; counts preview before sending.

7.2 Campaign automation

- Channels: **email + SMS** in v1 (Resend/Postmark for email, Twilio for SMS — final vendor choice at Phase 3 kickoff).
- Campaign types: one-off blasts, and **automations** triggered by events/conditions: welcome series, lapsed-guest win-back (churn-risk triggered), birthday/occasion, post-visit feedback ask, reward-earned notification.
- Personalisation: merge fields + AI-drafted copy variants per segment (venue approves before send; nothing auto-sends unreviewed copy in v1).
- Attribution: per-campaign UTM/short-link + redemption codes; dashboard reports opens, clicks, visits/orders attributed, revenue attributed.
- Compliance: per-channel consent enforced at send time; one-click unsubscribe; suppression lists; PECR-compliant sender identities.

7.3 Loyalty engine

- Configurable per venue (or shared across a group — TEK's multi-venue selling point): points per £ spent, visit stamps, or tiered (Bronze/Silver/Gold).
- Rewards: free item, tier perks, occasion bonuses. **No percentage-discount mechanics required for v1** — reward catalogue is value-add oriented.
- Guest experience: join in one step from any capture surface; balance visible on menu page; redemption via short code shown to staff (staff confirm in dashboard).
- AI assist: reward suggestions per segment based on preference data ("Top reward for your Friday regulars: free dessert — 68% of them order dessert").

7.4 Acceptance criteria (Module 3)

- A venue can create a segment, launch an automated win-back campaign, and see attributed redemptions end-to-end.
- Consent revocation stops all sends to that guest within minutes.
- A guest can join loyalty, earn on an order, see balance, and redeem in-venue with staff confirmation.
- Every send is logged (who, what, when, channel, consent basis).

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8. Module 4 — AI Predictive Analytics

8.1 Capabilities

Capability	Description	Method (v1)
Churn risk	Per-guest lapse probability, banded (healthy / cooling / at-risk / lost)	Gradient-boosted classifier on recency/frequency/monetary + engagement features; retrained weekly per tenant cohort
Demand forecast	Expected orders/covers by day and daypart, 14-day horizon	Time-series model (seasonality + trend + holiday calendar); per venue
Menu performance	Dish-level views → orders conversion, attach rate, margin contribution, rising/falling movers	Deterministic analytics over the event spine
Preference trends	Shifting taste-tag demand per venue (“vegan mains +22% this quarter”)	Aggregated tag analytics with significance thresholds

8.2 Dashboard

- Venue home: today’s expected demand, at-risk guest count with one-click “send win-back”, top movers on the menu, campaign performance summary.
- Weekly AI digest: plain-English summary of what changed and the single highest-impact suggested action (LLM-written from computed metrics only — every number in the digest traces to a stored metric).
- All predictions display confidence/coverage honestly; models degrade gracefully to heuristics when a venue has thin data, and the UI says so.

8.3 Acceptance criteria (Module 4)

- Churn bands populate for any venue with ≥ 90 days of data; backtest report generated per model version.
- Forecast accuracy tracked and displayed (MAPE vs actuals) — no unmeasured claims.
- Weekly digest delivered to venue owner by email; every figure reproducible from the dashboard.

8A. Module 5 — TEK Guest Mobile App (iOS & Android)

8A.1 Scope

Single cross-platform codebase (React Native/Expo) shipping to both stores under the TEK brand, consuming the same APIs as the web surfaces:

- **Venue entry:** scan a table QR or enter a venue code; recent venues remembered.
- **Menu & ordering:** full menu with allergen labels, dish detail, capture-only ordering — identical scope to the web ordering surface (§5.2).
- **For-you recommendations:** the Module 2 rail, with the same code-enforced allergen filtering.
- **Loyalty wallet:** balance/progress, reward catalogue, redemption short codes; join in one step.
- **Push notifications:** campaign and reward messages via a dedicated `push` consent channel (opt-in at first run, one-tap opt-out, enforced at send time exactly like email/SMS — Module 3’s compliance rules apply unchanged).
- **Account & privacy:** profile, consent management, data export/erasure — parity with the web privacy hub (§5.3, Module 1 GDPR tooling).

8A.2 Acceptance criteria (Module 5)

- App approved and live on both the Apple App Store and Google Play under TEK's accounts.
- A guest can scan a venue QR from the app, order, join loyalty, and redeem — end-to-end parity with the web journey.
- Push messages deliver only to guests with explicit push consent; opt-out stops sends within minutes.
- All allergen and consent guarantees hold identically to the web surfaces (shared API enforcement, verified by the same test suites).

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9. Architecture & Technical Design

9.1 Approach: modular monolith (approved)

One Next.js (App Router) application serving venue dashboard, guest capture surfaces, and API routes; PostgreSQL as the single source of truth with **row-level security keyed on tenant ID**; a Python worker service for ML jobs; LLM APIs for language tasks. Chosen over microservices (2–3× cost, unneeded at this scale) and third-party CRM assembly (per-contact fees destroy the £249/venue margin; TEK must own the intelligence layer).

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Guests (QR menu / ordering / signup)     Venue staff (dashboard)      TEK admin
         │                                       │                         │
         ▼                                       ▼                         ▼
┌──────────────────────────────────────────────────────────────────────────┐
│  Next.js app (Vercel) — guest surfaces · dashboard · admin · API routes  │
└──────────────────────────────────────────────────────────────────────────┘
                │                                          │
                │                                          │
                ▼                                          ▼
     PostgreSQL (managed, RLS multi-tenant)          Job queue (Postgres-backed)
     profiles · events · menus · segments                  │
     campaigns · loyalty · predictions                     ▼
                ▲                                  Python ML worker (VM/container)
                │                                  churn · forecast · taxonomy
                └── LLM API (OpenAI) ── explanations · copy drafts · digests
     Email (Resend) · SMS (Twilio) · Object storage (dish photos)
```

9.2 Stack

Layer	Choice	Rationale
Web/app/API	Next.js 15+, TypeScript, App Router	Matches TEK's advertised stack; one deploy target
Database	Managed PostgreSQL with RLS	Tenant isolation enforced at the database, not just app code
ML worker	Python 3.12 (scikit-learn, statsmodels/prophet-class), containerised	Right tool for churn/forecast; isolated from web path
Queue/jobs	Postgres-backed queue (e.g. pg-boss)	No extra infrastructure at this scale
LLM	OpenAI API (per TEK stack); provider-abstracted client	Explanations, copy drafts, taxonomy tagging, digests
Email / SMS	Resend or Postmark / Twilio	Deliverability + UK SMS compliance
Hosting	Vercel (app) + one small container host (worker) + managed Postgres	Minimal ops surface
Auth	Email/password + magic link; seat-based RBAC (Owner/Manager/Staff/TEK-Admin)	No social-login dependency

9.3 Data model (core tables, abbreviated)

`organizations`, `venues`, `seats/users`, `guests`, `guest_identities`, `consents`, `events` (partitioned, append-only), `menus`, `dishes`, `dish_tags`, `preferences` (declared/derived, provenance), `segments`, `campaigns`, `sends`, `loyalty_programs`, `loyalty_accounts`, `loyalty_transactions`, `rewards`, `redemptions`, `predictions` (model, version, score, band, computed_at), `model_runs` (metrics, backtests).

9.4 Security & non-functional requirements

- **Tenant isolation:** Postgres RLS on every tenant-scoped table; cross-tenant access is a test-suite failure class of its own.
- **PII:** encrypted at rest (managed-DB level) and in transit; PII columns access-audited; secrets in environment config, never in code.
- **GDPR/UK DPA:** consent audit trail, subject access export (JSON/CSV), erasure workflow (hard-delete PII, pseudonymise events), data-processing register documentation handed to TEK.
- **Performance:** guest surfaces LCP < 2.5 s on 4G; recommendation API < 300 ms p95; dashboard queries < 1 s p95.
- **Availability:** target 99.9% on managed platforms; graceful degradation — if ML worker or LLM is down, menus/ordering/loyalty continue (recommendations fall back to popularity, digests skip a week).
- **Observability:** structured logs, error tracking (Sentry-class), per-tenant usage metering (supports TEK's plan enforcement).
- **Testing:** unit + integration suites per module; the allergen-filter and consent-enforcement paths get exhaustive automated tests; each phase ends with a UAT script executed with TEK.

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10. Delivery Plan & Timeline

Phase	Contents	Duration	Demo/exit criteria
1 — Data Foundation	Multi-tenant core, auth/RBAC, venue onboarding, menu manager + AI taxonomy tagging, QR menu + ordering + signup surfaces, guest profiles + event spine, GDPR tooling, TEK admin panel	5 weeks	A pilot venue onboards itself, guests browse/order/sign up, profiles populate, erasure works
2 — Intelligence	Recommendation engine (all 3 stages), analytics dashboard, churn model, demand forecast, menu performance, weekly digest	5 weeks	Live recommendations on the pilot venue's menu; churn bands + forecasts on real captured data; backtest reports
3 — Activation	Segment builder, campaign automation (email+SMS), attribution, loyalty engine + guest surfaces + staff redemption, AI copy/reward assist	5 weeks	End-to-end: segment → automated campaign → attributed redemption; loyalty earn/redeem live
4 — POS Integrations	Square, Toast or Lightspeed connectors (2 in scope): historical import + ongoing order sync into the event spine	4 weeks	Pilot venue's POS orders flow into profiles and retrain models
5 — Mobile App	TEK Guest app (iOS & Android): menu/ordering, recommendations, loyalty wallet, push notifications, privacy parity	4 weeks (overlaps Phase 4)	App live in both stores; end-to-end guest journey passes on device

Total programme (Phases 1–5): **~19 weeks** from kickoff (Phases 4 and 5 run in parallel). Each phase gates on written acceptance before the next begins.

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11. Running Costs (TEK's accounts, estimates)

Item	Est. monthly (pilot scale, ≤ 25 venues)
Hosting (Vercel Pro + worker host + managed Postgres)	£90–£180
Email (per-send)	£15–£60
SMS (Twilio UK, usage-based)	£0.04/msg — campaign dependent
LLM API	£30–£120 (explanations cached; copy drafts on demand)
Error tracking / monitoring	£0–£25

Per-venue infrastructure cost at pilot scale: **≈ £6–£15/month**, comfortably inside the £249/venue price point.

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12. Commercial Terms — Agreed All-Inclusive Engagement

Terms agreed between TEK and OEFR Enterprise Inc in direct negotiation, superseding the v1.0 two-track quotation (OEFR-PF-2026-001/-002). One fixed price covers the complete programme — all five phases, nothing optional:

Phase	Fixed Price
1 — Data Foundation	£4,999
2 — Intelligence	£4,800
3 — Activation	£4,000
4 — POS Integrations (2 connectors)	£2,500
5 — Mobile App (iOS & Android)	£2,500
Programme total (all-inclusive)	**£18,799**

Delivery wrap: solo senior delivery, AI-accelerated; 30-day post-phase defect warranty; async support (48 h response); essential documentation (admin guide + API reference); dedicated staging environment with seeded demo tenant (already live for TEK review).

Ongoing operations (optional, post-launch)

Plan	Scope	Monthly
Ops Essential	Monitoring, patching, model retraining oversight, 8 h/month improvements	£1,500
Ops Growth	Essential + 24 h/month feature development, campaign/model tuning, monthly performance report	£3,500

Payment terms

- **40% to commence (£7,519), 60% on final written acceptance (£11,280)** of the Phase 5 exit criteria.
- Prices exclude VAT (if applicable) and third-party running costs (Section 11, plus app store developer fees on TEK's accounts).
- Terms valid 30 days from the date of this document.
- Scope changes handled by written change order against a day rate agreed at signature.

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13. Risks & Mitigations

Risk	Mitigation
Thin data at pilot venues weakens models	Heuristic fallbacks shipped first; models activate at data thresholds; honest confidence display
Email/SMS deliverability	Dedicated sending domains, warm-up plan, suppression hygiene from day one
Allergen/dietary errors	Hard-coded constraint filtering (never model-decided), exhaustive automated tests, venue confirmation step on taxonomy
GDPR exposure	Consent-first design, erasure workflow tested in UAT, processing register delivered
Scope drift from TEK's broader vision (VR, hardware)	Fixed per-phase scope with written acceptance gates; change-order process
LLM cost/latency	Explanations cached per (guest-segment × dish); language tasks off the critical path

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14. Acceptance & Next Steps

1. TEK confirms written acceptance of this document and the agreed terms (pro forma invoice OEFR-PF-2026-003).
2. Statement of Work + commencement invoice issued; kickoff within 5 business days of deposit.
3. Weekly written progress reports; phase demos on the shared staging environment (already live).

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